SCOOT LEARNING LAB · STUDY GUIDE

The traffic light that sees the queue before it forms

How SCOOT counts cars far up the road, predicts the queue at the lights, and nudges the timings a few seconds at a time.

Level 1 · Foundation



INSIDE

- Why a loop 200 m from the lights is more useful than one at the lights
- The 'link profile unit': why a bus counts as three cars
- How a moving platoon spreads out, worked by hand
- A bathtub model of the queue, and the 90% rule

Beginner-friendly companion to smrahmanbd.github.io/scoot-learning-lab

The map: SCOOT Foundation

This guide is for complete beginners. You need no traffic engineering and only school maths. By the end you will understand how SCOOT, one of the world's most widely used adaptive traffic control systems, builds a live model of every road and uses it to time the lights. There are six ideas. Each builds on the one before.

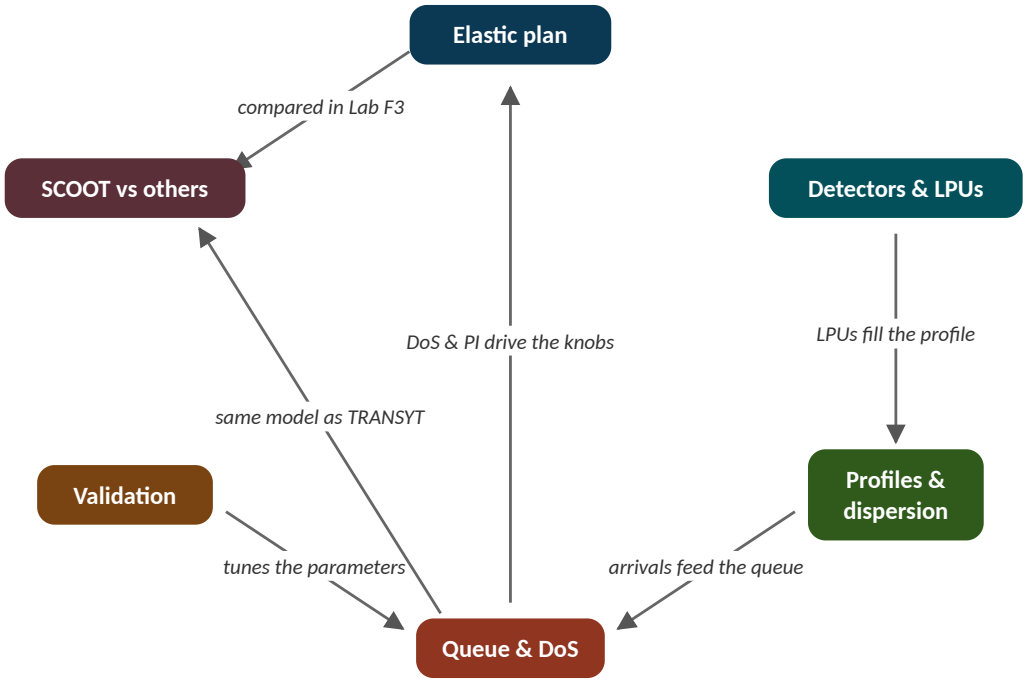


Figure 0. The whole guide on one page. Each box is one concept section; the arrows say how one idea feeds the next. Come back to this page whenever you feel lost.

1. What SCOOT is: the elastic plan
2. Detectors and link profile units
3. Profiles and platoon dispersion
4. The queue model and degree of saturation
5. Validation: making the model match the street
6. SCOOT against fixed time and actuation

HOW TO USE THIS GUIDE

Read one concept per sitting (20 to 30 minutes). Do every **Simple calculation** with a pencil before reading the answer. Say the **Picture this** scene out loud: the sillier it feels, the better it sticks. The **Recap so far** box at the end of each section grows as you go; reading it is your free revision. When you finish, take the self-quiz, then follow the revision plan on the last page.

What you will be able to do

- Explain in two sentences what SCOOT changes (split, cycle, offset) and why it changes them in small steps.
- Turn a vehicle's length into link profile units (LPUs) and explain why long vehicles count more.
- Shift and smooth a platoon from the detector to the stop line with Robertson's formula.
- Draw the queue over one cycle and read off the delay, stops and degree of saturation.
- Say what an engineer 'validates' and fix a wrong saturation occupancy from a stopwatch reading.
- Explain why SCOOT beat fixed time and actuation in the SUMO lab, and where fixed plans come from (Webster).

Before we start

HOOK

The detector is in the 'wrong' place on purpose. Most actuated signals put their loops at the stop line. SCOOT puts them near the far end of the road, just after the previous junction. That gives it 10 to 20 seconds of warning: it knows a platoon is coming before any driver reaches the lights.

HOOK

SCOOT never sees the queue. There is no camera watching the queue. Every queue SCOOT uses is a prediction from its model. In our SUMO lab the model's queue still tracked the real queue closely, second by second.

HOOK

It has been running since about 1980. SCOOT was developed in the UK by the Transport and Road Research Laboratory (now TRL) and is used in hundreds of towns and cities. SCOOT 7 (2020) added connected-vehicle data and pedestrian features.

Words you will meet

You do not need to memorise these now. Each one is explained again when it first matters.

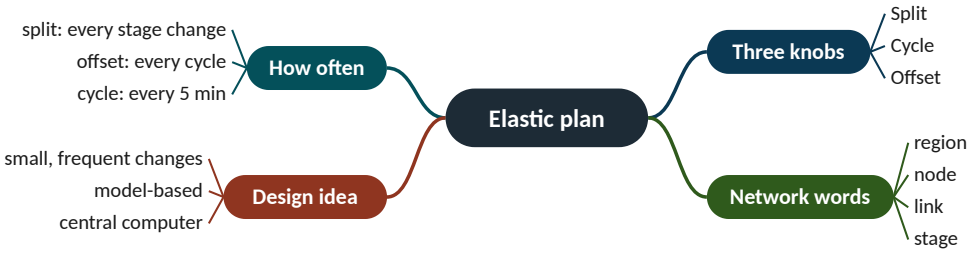
Word	In plain words
Junction / node	A place where roads meet and traffic lights take turns. SCOOT calls each one a **node**.
Link	One approach road into a junction, from the detector to the stop line.
Stage	A set of movements that get green at the same time, for example 'main road both ways'.
Cycle	One full turn through all stages, for example 60 seconds.
Split	How the cycle is shared between stages: 'main road 35 s, side road 20 s'.
Offset	When one junction's cycle starts compared with its neighbour, so a green wave can form.
Platoon	A bunch of cars travelling together, usually released by the previous green.
Detector (loop)	A wire loop in the road that knows when a vehicle is over it.
Saturation flow	The fastest a queue can leave on green, roughly one car every two seconds per lane.

CONCEPT 1 OF 6 · SCOOT FOUNDATION

What SCOOT is: the elastic plan

HOOK

A fixed-time traffic light runs the same plan whether the road is empty or jammed. SCOOT changes the plan all day long, but never by much at once. Why would small steps beat big jumps?



IN PLAIN WORDS

Think of a thermostat that nudges the heating a little every few minutes, rather than a timer that switches between 'winter' and 'summer' settings. Each nudge is small, so one wrong guess cannot do much harm, and the many nudges add up to follow the weather closely. *Where the picture breaks:* a thermostat has one knob; SCOOT turns three knobs, each at its own speed, for every junction in an area.

SCOOT stands for **Split Cycle Offset Optimisation Technique**. The name lists the three timing 'knobs' it turns. The **split** is how the green time is shared between stages. The **cycle** is how long one full round of stages takes. The **offset** is when a junction's cycle starts compared with its neighbour.

SCOOT is **centralised** and **model-based**. One computer in the control centre runs a small traffic model for every approach road (link) every second. The model's predictions, not the raw detector counts, decide the timings.

Each knob turns at its own speed. Splits change a few seconds before every stage change. Offsets change once per cycle. The cycle, which is shared by all junctions in a **region**, changes every 5 minutes (every 2.5 minutes while traffic is rising). This is the **elastic plan**: the timings stretch to fit demand instead of jumping between stored plans.

SIMPLE CALCULATION · How many decisions in one hour?

One junction, two stages, a 60 s cycle, in a region whose cycle is reviewed every 5 minutes.

1. Cycles per hour = 3600 ÷ 60 = **60**.
2. Split decisions: one before each stage change, two changes per cycle, so 60 × 2 = **120**.
3. Offset decisions: one per cycle, so **60**.
4. Cycle decisions: one per 5 minutes, so 60 ÷ 5 = **12**.

About 192 small decisions an hour at one junction. That is why each one can be small: there are plenty more to come.

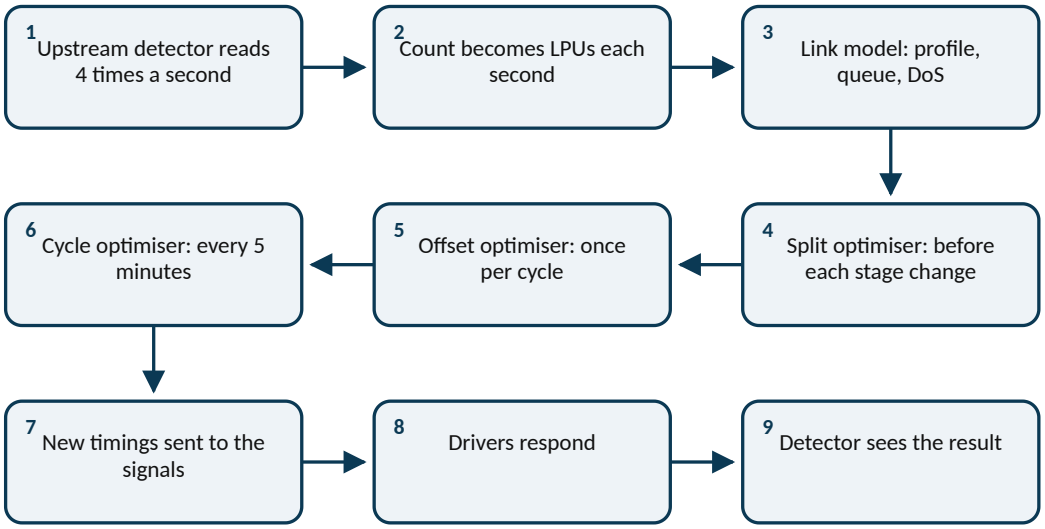


Figure 1. The SCOOT loop. What to notice: the detector feeds the model, the model feeds three optimisers working at three speeds, and the drivers' response is measured again by the detector. Read it left to right, then down.

TRAP

SCOOT does **not** pick a new plan from a library of plans (that is plan selection, an older idea). It edits the plan it is already running, a few seconds at a time.

PICTURE THIS · STOP 1 OF THE ROUTE

At the town gate a giant rubber band is stretched around the traffic light. Three dials on the pole are labelled S, C and O. A tiny engineer twangs the band every few seconds; it stretches, never snaps. *What it encodes:* The rubber band is the elastic plan; the dials are split, cycle and offset; the frequent twangs are small, frequent changes.

TRY IT NOW (about 90 seconds)

A junction has three stages and an 80 s cycle. How many split decisions does SCOOT make in an hour? *Answer in the key at the back.*

RECAP SO FAR

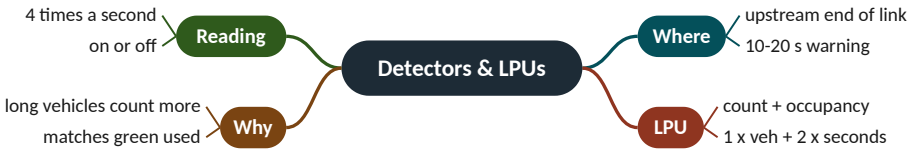
1. **Elastic plan: SCOOT turns three knobs (split every stage change, offset every cycle, cycle every 5 min) by small amounts, using a model.**

CONCEPT 2 OF 6 · SCOOT FOUNDATION

Detectors and link profile units

HOOK

A bus takes up more green time than a car. How could a simple wire loop in the road know that, without knowing what a bus is?



IN PLAIN WORDS

Imagine a doorman who charges every person one coin to enter, plus two coins for every second they block the door. A quick walker pays a little over one coin; someone pushing a wide trolley pays three. The bill measures how much of the doorway each person used, not just how many came in. *Where the picture breaks:* a real loop also 'charges' for slow traffic, because a slow car stays on the loop longer, which is actually useful: slow traffic usually means queueing.

A loop detector is a coil of wire under the road surface. When metal passes over it, it switches 'on'. SCOOT reads every loop four times a second, so it knows both **how many** vehicles crossed and **how long** the loop was covered (the **occupancy**).

SCOOT places its loops at the **upstream** end of each link, just after the previous junction. A vehicle crossing there will reach the stop line 10 to 20 seconds later, which gives the model time to act.

SCOOT turns each second of loop data into **link profile units (LPUs)**, a blend of count and occupancy. Long vehicles cover the loop for longer, so they produce more LPUs, just as they need more green to clear. TRL's exact coding is not published; this lab uses the simple rule below, which behaves the same way.

$$\text{occupancy time} = (\text{vehicle length} + \text{loop length}) \div \text{speed}$$

how long the loop is covered

$$\text{LPU} = 1 \times \text{vehicles} + 2 \times \text{seconds occupied}$$

this lab's definition (illustrative)

SIMPLE CALCULATION · A car and a bus at 13.5 m/s over a 2 m loop

Car length 4.5 m. Bus length 12 m. Speed 13.5 m/s (about 49 km/h).

1. Car occupancy: (4.5 + 2) ÷ 13.5 = 6.5 ÷ 13.5 = **0.48 s**.
2. Car LPU: 1 + 2 × 0.48 = **1.96 LPU**.
3. Bus occupancy: (12 + 2) ÷ 13.5 = 14 ÷ 13.5 = **1.04 s**.
4. Bus LPU: 1 + 2 × 1.04 = **3.07 LPU**.
5. Ratio: 3.07 ÷ 1.96 = **1.57**.

The bus counts as about one and a half cars, without the loop ever 'knowing' it was a bus. The SUMO validation lab measured 1.95 LPU per vehicle on average, very close to our car.

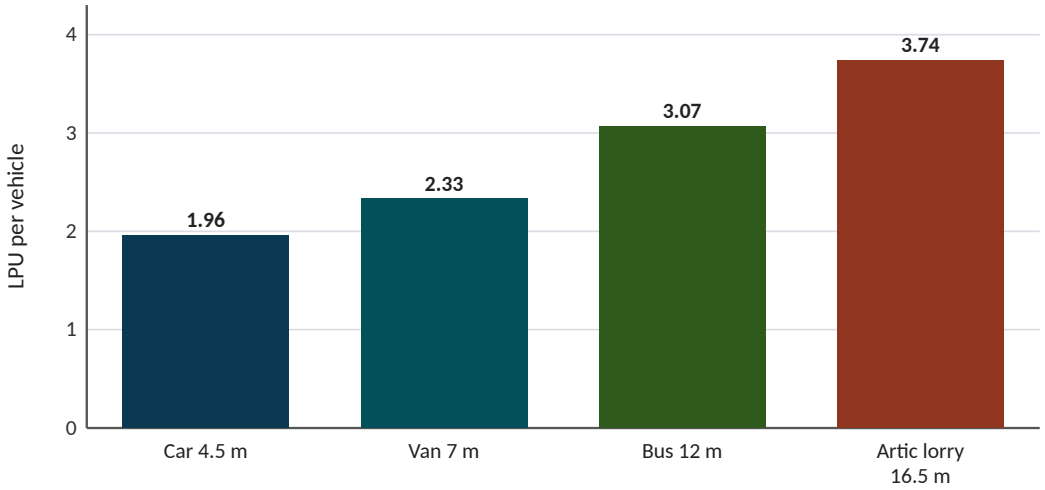


Figure 2. LPUs per vehicle at 13.5 m/s. What to notice: LPUs grow with length, so the model 'weights' each vehicle by the green it will use. Computed with the formula above (2 m loop).

TRAP

Do not think LPUs are 'vehicles'. A queue of 10 cars is about 20 LPUs in this lab. Always check whether a number is in vehicles or LPUs before you compare it.

PICTURE THIS · STOP 2 OF THE ROUTE

Just past the gate, a doormat across the whole road is a cash register. Every vehicle pays ONE gold coin to roll on, then TWO coins for every second it sits there. A long red bus groans, paying three coins and dropping one on its foot.

What it encodes: LPU = 1 × vehicles + 2 × seconds occupied; the bus costs about 3 LPUs, a car about 2. The mat is far from the lights: the upstream detector.

TRY IT NOW (about 90 seconds)

A 16.5 m lorry crosses the 2 m loop at 9.25 m/s. How many LPUs does it produce? *Answer in the key at the back.*

RECAP SO FAR

1. **Elastic plan:** SCOOT turns three knobs (split every stage change, offset every cycle, cycle every 5 min) by small amounts, using a model.
2. **Detectors and LPUs: loops sit upstream (10-20 s warning); LPU = count + occupancy, so long or slow vehicles count more.**

CONCEPT 3 OF 6 · SCOOT FOUNDATION

Profiles and platoon dispersion

HOOK

Cars leave a green light as a tight bunch. By the next junction the bunch has stretched out. Can you predict exactly how much, with one line of arithmetic?



IN PLAIN WORDS

Picture a school class leaving the gate together and walking to the bus stop. The fast walkers pull ahead, the slow ones fall behind. At the bus stop the class arrives over a longer time than it took to leave, and the busiest moment is less busy. *Where the picture breaks:* cars cannot overtake freely on a busy road, so real dispersion depends on the road. The model uses fixed constants that engineers can tune.

Each second SCOOT adds the new LPUs into a **cyclic flow profile**: a bar chart of flow against 'second of the cycle'. Because the signals repeat every cycle, the profile shows where in the cycle platoons arrive. It is smoothed over several cycles (new value = 0.75 × old + 0.25 × new) so one odd cycle does not dominate.

To know when traffic reaches the stop line, SCOOT moves the profile forward by about the **cruise time** T (detector to stop line at cruising speed) and spreads it out. It uses **Robertson's dispersion model**, the same one used by the TRANSYT design program.

Read the formula like a recipe: each second's arrival at the stop line is a fraction F of what crossed the detector, plus the rest (1 − F) of the previous second's arrival. A small F means a lot of spreading. Longer roads (bigger T) give smaller F.

$$q_{stop}(t + \beta T) = F \cdot q_{det}(t) + (1 - F) \cdot q_{stop}(t + \beta T - 1)$$

flow at the stop line, one second at a time

$$F = 1 \div (1 + \alpha\ \beta\ T),\ \alpha = 0.35,\ \beta = 0.8$$

T = cruise time in seconds; the lag is βT

SIMPLE CALCULATION · A 10-second platoon on a 10-second link

A platoon crosses the detector at 1 LPU per second for 10 seconds. Cruise time T = 10 s.

1. Lag βT = 0.8 × 10 = **8 s**: the first arrivals appear 8 s after the first detection.
2. F = 1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 = **0.263**.
3. Second 1 at the stop line: 0.263 × 1 + 0.737 × 0 = **0.263**.
4. Second 2: 0.263 × 1 + 0.737 × 0.263 = **0.457**.
5. Second 3: 0.263 × 1 + 0.737 × 0.457 = **0.600**.
6. Keep going: the arrivals reach a peak of 0.95 after 10 s and then fade away over roughly 15 more seconds.

The same 10 LPUs arrive, but spread over about 25 s with a lower peak (0.95 instead of 1.0). Nothing is lost; it is only rearranged in time.

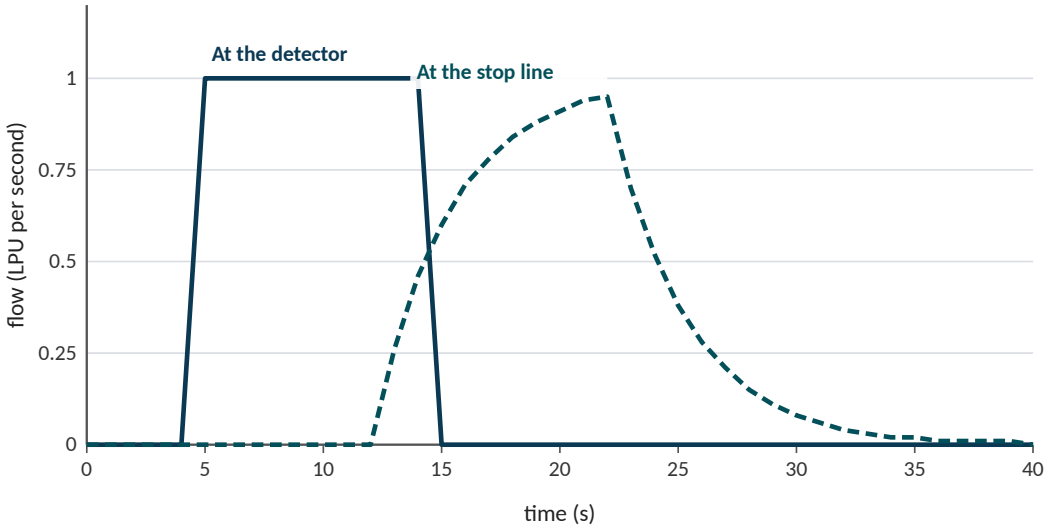


Figure 3. The same platoon at the detector and at the stop line. What to notice: the stop-line curve starts 8 s later, rises gradually, and has a long tail. The areas under the two curves are equal (10 LPU). Computed with Robertson's formula, T = 10 s.

TRAP

The lag is not the full cruise time. In Robertson's model the platoon's front arrives at βT = 0.8 T, because the fastest drivers get there early. The rest arrive later in the tail.

PICTURE THIS · STOP 3 OF THE ROUTE

On a bridge, a marching band of 35 players leaves in a tight block. A sign says 'walk at 80% speed'. By the far end of the bridge the trombones are way ahead and the drummers trail far behind, still playing. *What it encodes:* 35 players and 80% encode α = 0.35 and β = 0.8; the band spreading out is dispersion; the far end of the bridge is the stop line.

TRY IT NOW (about 90 seconds)

Find F and the lag for a long link with cruise time T = 20 s. Will the platoon spread more or less than in the worked example? *Answer in the key at the back.*

RECAP SO FAR

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3. **Profiles and dispersion: LPUs build a smoothed cyclic profile; Robertson shifts it by 0.8T and spreads it with F = 1/(1 + 0.35·0.8·T).**

CONCEPT 4 OF 6 · SCOOT FOUNDATION

The queue model and degree of saturation

HOOK

If you know when cars arrive and when the light is green, you can draw the queue without ever seeing it. SCOOT does exactly that, for every link, every second.



IN PLAIN WORDS

A queue is a bathtub. Arrivals are the tap, always running. During red the plug is in, so the water rises. During green the plug is out and the water drains at a fixed speed (the saturation occupancy). If the tap is slower than the drain, the tub empties during green. *Where the picture breaks:* a real queue has a length on the road and can block the junction behind it; the basic bathtub ignores that until the Advanced level (congestion).

From the predicted arrivals at the stop line, the model follows simple rules. Arrivals during red join the queue. During **effective green** the queue leaves at the **saturation occupancy (SATO)**, in LPUs per second. Effective green starts a **start lag** after the light turns green (drivers react) and ends an **end lag** after it turns amber (drivers keep going through amber).

From the queue the model reads three things. **Delay** is the area under the queue curve (LPU·seconds). **Stops** are arrivals that meet a red light or a queue. The **degree of saturation (DoS)** is how full the green is: demand in the cycle, plus any queue left over, divided by what the green could discharge.

SCOOT tries to keep every link's DoS at or below about **90%**, leaving a margin for random surges. The optimisers compare options using a **performance index (PI)**: delay plus a weight K times stops.

$$\text{DoS} = (\text{LPU arriving per cycle} + \text{residual queue}) \div (\text{effective green} \times \text{SATO})$$

1.0 means the green is exactly full

$$\text{PI} = \text{delay} + \text{K} \times \text{stops}$$

lower is better; K sets how much a stop matters

SIMPLE CALCULATION · One 60 s cycle on one link

Arrivals 0.2 LPU/s, steady. Effective red 35 s, effective green 25 s. SATO 0.6 LPU/s. No queue at the start.

1. Queue at the end of red: 0.2 × 35 = **7 LPU**.
2. In green the queue shrinks at 0.6 − 0.2 = 0.4 LPU/s, so it clears after 7 ÷ 0.4 = **17.5 s**.
3. Delay = triangle areas: ½ × 35 × 7 + ½ × 17.5 × 7 = 122.5 + 61.25 = **183.75 LPU·s** per cycle.
4. Stops = everyone who arrived before the queue cleared: 0.2 × (35 + 17.5) = **10.5 LPU**.
5. Demand per cycle = 0.2 × 60 = 12 LPU. Capacity = 25 × 0.6 = 15 LPU. DoS = 12 ÷ 15 = **0.80**.

DoS is 80%, under the 90% target, so this link has some spare green to give away.

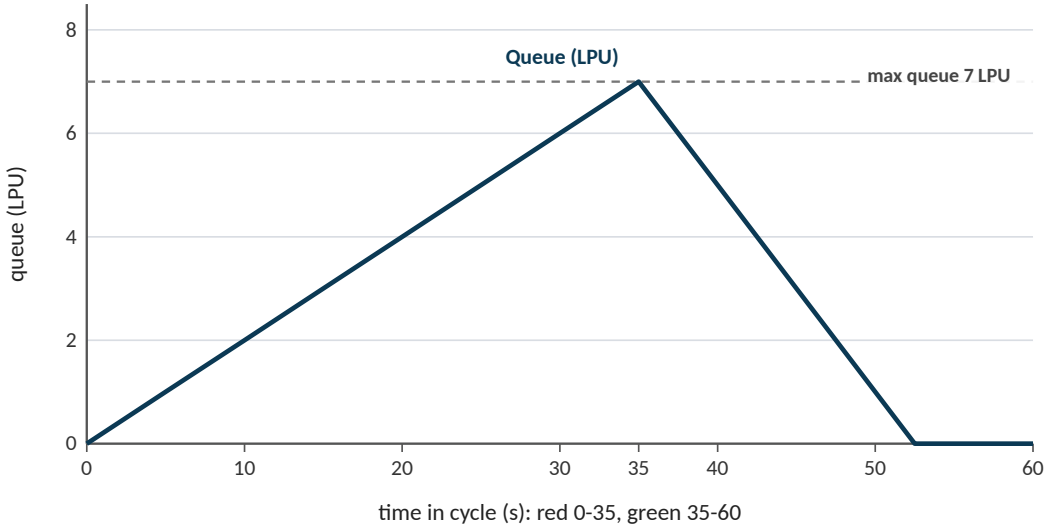


Figure 4. The modelled queue over one cycle. What to notice: the queue grows in a straight line during red, falls faster during green, and is empty for the last 7.5 s of green. Delay is the area under the curve. Values from the worked example.

TRAP

A DoS of 100% does not mean the road is jammed forever; it means the green is exactly used up. Above 100% a residual queue carries into the next cycle and grows. Also: DoS uses **effective** green, not the displayed green.

PICTURE THIS · STOP 4 OF THE ROUTE

At the stop line stands a huge bathtub with a tap that never stops. When the light is red, a rubber plug shaped like a red light is in. On green the plug pops and water swirls out. A gauge on the side is painted red above the 90% mark.

What it encodes: Tap = arrivals; drain speed = SATO; red plug = red; the 90% mark = SCOOT's DoS target; the water's area over time = delay.

TRY IT NOW (about 90 seconds)

Same link, but arrivals rise to 0.24 LPU/s. What is the DoS? Is it under 90%? *Answer in the key at the back.*

RECAP SO FAR

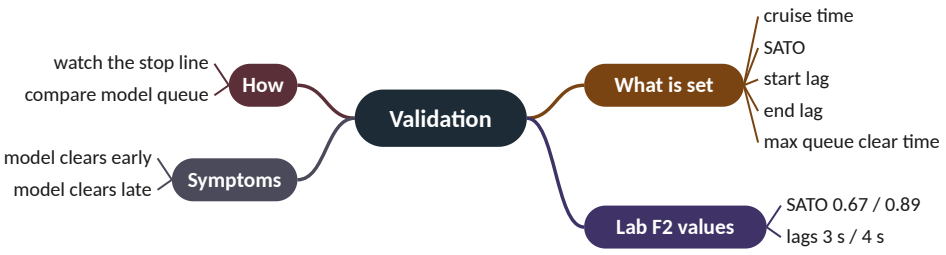
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4. **Queue and DoS: red fills the 'bathtub', green drains it at SATO; delay = area, DoS = demand ÷ (green × SATO), target ≤ 90%; PI = delay + K·stops.**

CONCEPT 5 OF 6 · SCOOT FOUNDATION

Validation: making the model match the street

HOOK

Poor validation is the most common reason SCOOT underperforms. The cure is low-tech: an engineer at the kerb with a stopwatch.



IN PLAIN WORDS

Validation is like tuning a guitar by ear. The model is the guitar; the street is the tuning fork. You pluck, listen, and turn the peg until they agree. A perfectly built guitar still plays wrong notes if nobody tunes it. *Where the picture breaks:* a guitar stays in tune for weeks; a link may need re-validating whenever the road layout, lanes or parking change.

The model has a few numbers that depend on the actual street: the **cruise time** from detector to stop line, the **saturation occupancy (SATO)** at which a queue discharges, the **start lag** and **end lag**, and the **maximum queue clear time**. When a link is commissioned, an engineer stands at the stop line and compares what they see with what SCOOT shows.

The simplest check: when does the queue clear? If the model's queue empties earlier than the real one, the model thinks traffic leaves faster than it does, so SATO is too high. If it empties later, SATO is too low.

In this lab's SUMO network, Lab F2 measured 1.95 LPU per vehicle, cruise speed 13.5 m/s, SATO 0.67 LPU/s per lane on the main road and 0.89 on the side road, a start lag of 3 s and an end lag of 4 s. Those became the controller's defaults.

$$\text{queue clear time} = \text{start lag} + \text{queue} \div \text{SATO}$$

for a queue standing at the start of green

$$\text{SATO} = \text{queue} \div (\text{observed clear time} - \text{start lag})$$

rearranged to correct SATO from a stopwatch

SIMPLE CALCULATION · Fixing SATO from a stopwatch

A queue of 10 cars (about 19.5 LPU) is waiting. Model SATO 0.67 LPU/s, start lag 3 s. The engineer sees the queue clear 36 s after green starts.

1. Model clear time: 3 + 19.5 ÷ 0.67 = 3 + 29.1 = **32.1 s**.
2. Observed: **36 s**. The model clears about 4 s early, so its SATO is too high.
3. Corrected SATO = 19.5 ÷ (36 − 3) = 19.5 ÷ 33 = **0.59 LPU/s**.
4. Check: 3 + 19.5 ÷ 0.59 = 3 + 33.0 = **36.0 s**. It now matches.

A 12% change in SATO moved the model's clear time by 4 s. Small parameter errors give big errors in delay and DoS.

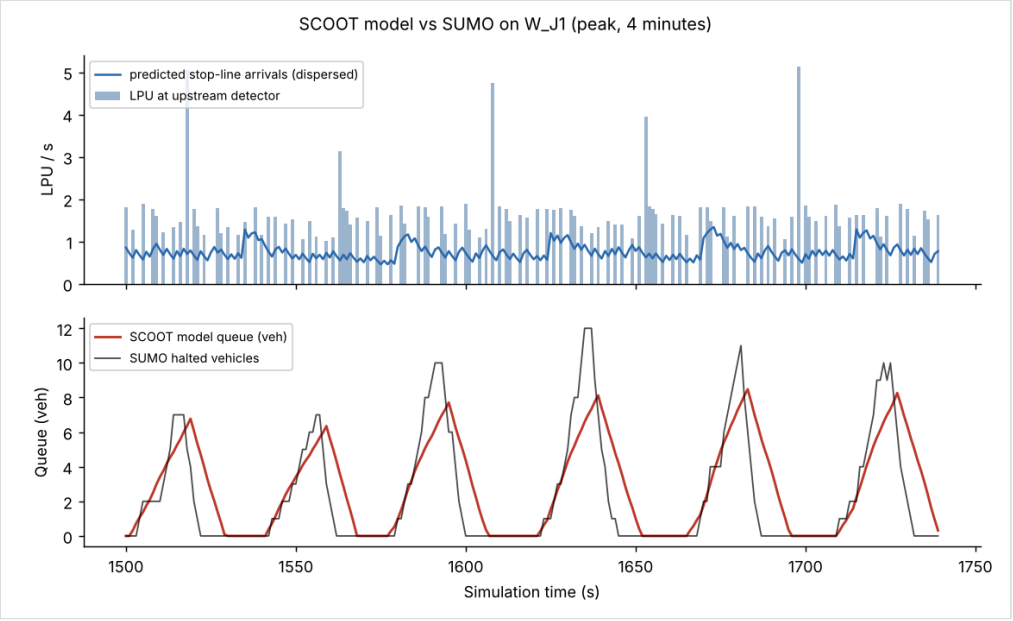


Figure 5. SCOOT's model queue against SUMO's real queue (Lab F1). What to notice: the model never sees the real queue, yet it follows it in timing and size. The model empties a few seconds later because SUMO stops counting a car as queued the moment it starts moving.

TRAP

Do not 'fix' a model by changing the target DoS or the optimiser. If the queue picture is wrong, every decision built on it is wrong. Validate the model first.

PICTURE THIS · STOP 5 OF THE ROUTE

Beside the stop line a guitar shop has a giant guitar strung across the road. An engineer with a stopwatch in her teeth twists a tuning peg marked SATO. The peg has two smaller pegs labelled 3 and 4.

What it encodes: Tuning = validation; the SATO peg = saturation occupancy; 3 and 4 = start lag 3 s and end lag 4 s; the stopwatch = observed queue clear time.

TRY IT NOW (about 90 seconds)

The model's queue clears at 30 s, but on the street it clears at 27 s (queue 18 LPU, start lag 3 s). Is SATO too high or too low? What should it be? *Answer in the key at the back.*

RECAP SO FAR

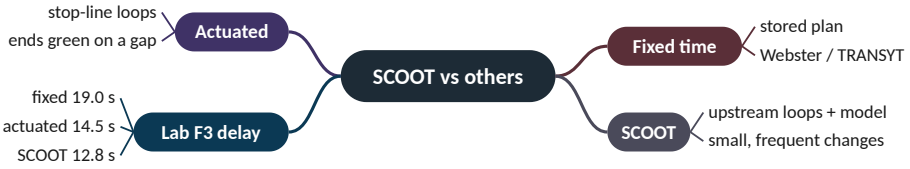
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4. **Queue and DoS:** red fills the 'bathtub', green drains it at SATO; delay = area, DoS = demand ÷ (green × SATO), target ≤ 90%; PI = delay + K·stops.
5. **Validation: tune cruise time, SATO, start/end lag and max queue clear time so the model's queue clears when the real one does.**

CONCEPT 6 OF 6 · SCOOT FOUNDATION

SCOOT against fixed time and actuation

HOOK

On one isolated junction, a simple actuated signal is already good. So how much does SCOOT add, and where does it really shine?



IN PLAIN WORDS

Fixed time is a printed bus timetable. Actuation is a driver who leaves as soon as nobody else is running for the bus. SCOOT is a dispatcher with a radio who hears that a crowd is coming in 15 seconds and plans for it. *Where the picture breaks:* at a single junction the dispatcher's advantage is small; it grows when many junctions must work together (the Advanced level).

Fixed-time signals repeat a stored plan, designed offline from traffic counts. For one junction the classic design is **Webster's formula**. For networks, engineers used TRANSYT, which uses the same profile and dispersion model as SCOOT. SCOOT is often called an 'on-line TRANSYT'.

Actuated signals have loops at the stop line and end a green when the gap between vehicles gets too long. They react well to random gaps but cannot see what is coming and do not coordinate.

In Lab F3 (one junction, one simulated hour in SUMO, same arrivals for all) mean delay was **18.96 s** for fixed time, **14.48 s** for actuation and **12.79 s** for SCOOT. SCOOT's cycle optimiser shortened the cycle in light traffic and held it while DoS rose in the peak.

$$C_0 = (1.5\,L + 5) \div (1 - Y)$$

Webster's optimum cycle: L = lost time per cycle, Y = sum of critical flow ratios q/s

$$g_i = (C_0 - L) \times y_i \div Y$$

effective green shared in proportion to flow ratio

SIMPLE CALCULATION · Webster for a two-stage junction

Lost time L = 10 s. Main road flow ratio y_A = 0.35, side road y_B = 0.25.

1. Y = 0.35 + 0.25 = **0.60**.
2. C_0 = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = **50 s**.
3. Effective green to share = 50 − 10 = **40 s**.
4. Main road: 40 × 0.35 ÷ 0.60 = **23.3 s**. Side road: 40 × 0.25 ÷ 0.60 = **16.7 s**.

A good fixed plan for these flows: 50 s cycle, 23.3 s and 16.7 s of effective green. It stays good only while the flows stay the same, which is why SCOOT adapts.

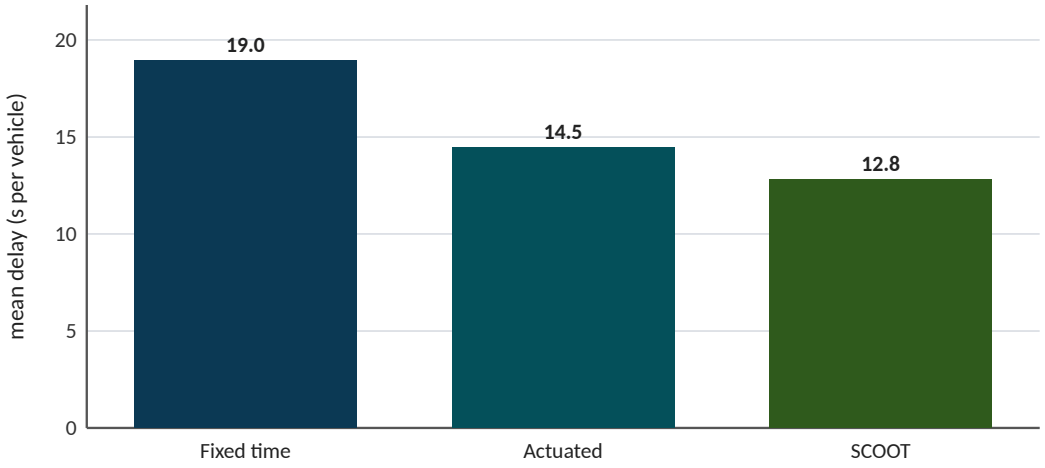


Figure 6. Mean delay per vehicle in Lab F3 (SUMO, one junction, one hour). What to notice: SCOOT beats fixed time by about a third and actuation by about 12%. The gap to actuation is small at an isolated junction; coordination is where SCOOT's model pays off.

TRAP

Webster's formula blows up as Y approaches 1: at Y = 0.9 and L = 10, C_0 = 200 s. Real controllers cap the cycle, and when Y ≥ 1 the junction is simply over capacity: no cycle length can fix it.

PICTURE THIS · STOP 6 OF THE ROUTE

Outside the control centre three runners finish a race. A robot with a printed timetable on its chest crosses at 19, a jumpy rabbit at 14 and a half, and a calm dispatcher wearing headphones at just under 13. On the podium sits a set of scales labelled '1.5 L + 5'.

What it encodes: Finish times are the Lab F3 delays (19.0, 14.5, 12.8 s); the scales are Webster's formula $C_0 = (1.5L + 5)/(1 - Y)$.

TRY IT NOW (about 90 seconds)

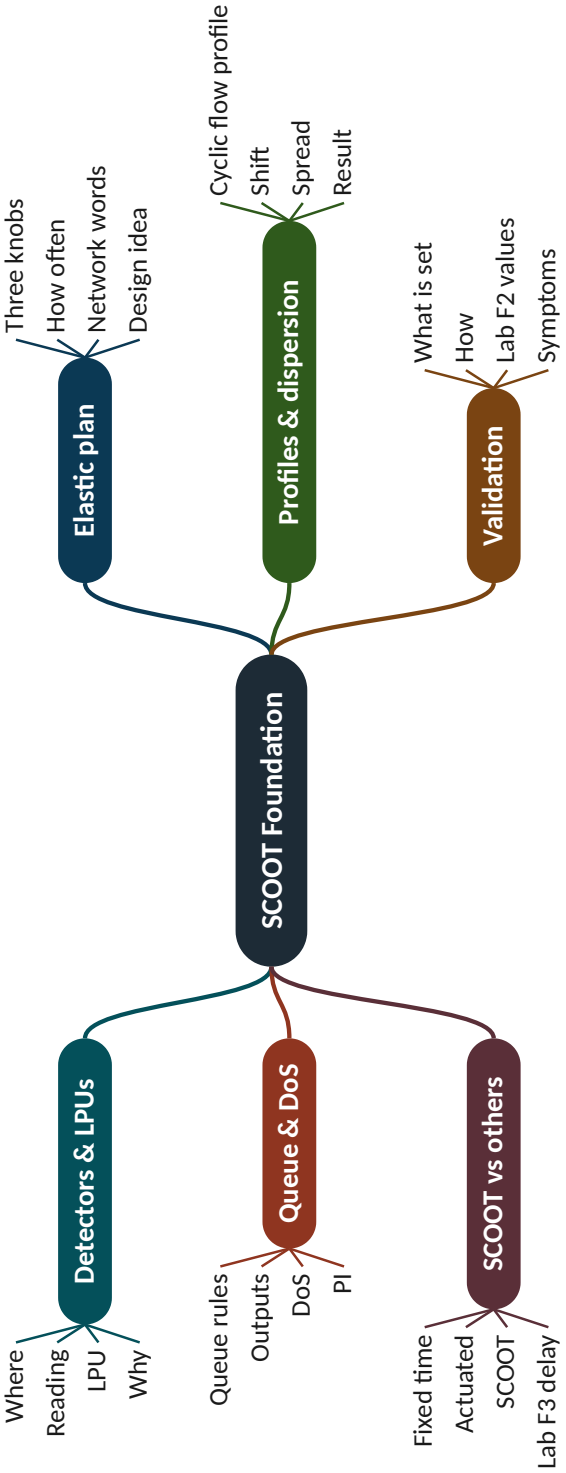
Use Webster with L = 12 s, y_A = 0.40, y_B = 0.30. Find C_0 and both greens. *Answer in the key at the back.*

RECAP SO FAR

1. **Elastic plan:** SCOOT turns three knobs (split every stage change, offset every cycle, cycle every 5 min) by small amounts, using a model.
2. **Detectors and LPUs:** loops sit upstream (10-20 s warning); LPU = count + occupancy, so long or slow vehicles count more.
3. **Profiles and dispersion:** LPUs build a smoothed cyclic profile; Robertson shifts it by 0.8T and spreads it with F = 1/(1 + 0.35·0.8·T).
4. **Queue and DoS:** red fills the 'bathtub', green drains it at SATO; delay = area, DoS = demand ÷ (green × SATO), target ≤ 90%; PI = delay + K·stops.
5. **Validation:** tune cruise time, SATO, start/end lag and max queue clear time so the model's queue clears when the real one does.
6. **SCOOT vs others: fixed plans come from Webster/TRANSYT; actuation reacts at the stop line; SCOOT predicts. Lab F3: 19.0 / 14.5 / 12.8 s delay.**

The master mind map

Every concept and its main branches on one page. Turn the page sideways. Try redrawing it from memory on blank paper, then compare.



Connections and contrasts

Three ways to run a traffic light

	Fixed time	Actuated	SCOOT
Detectors	none	at the stop line	upstream, 10-20 s before the stop line
Decides with	a stored plan	gaps between vehicles	a model of every link
Coordinates junctions?	yes, if designed so	no	yes, through offsets and a shared cycle
Reacts to change	no	every second	small steps: stage change, cycle, 5 min
Lab F3 delay	18.96 s	14.48 s	12.79 s

Things beginners mix up

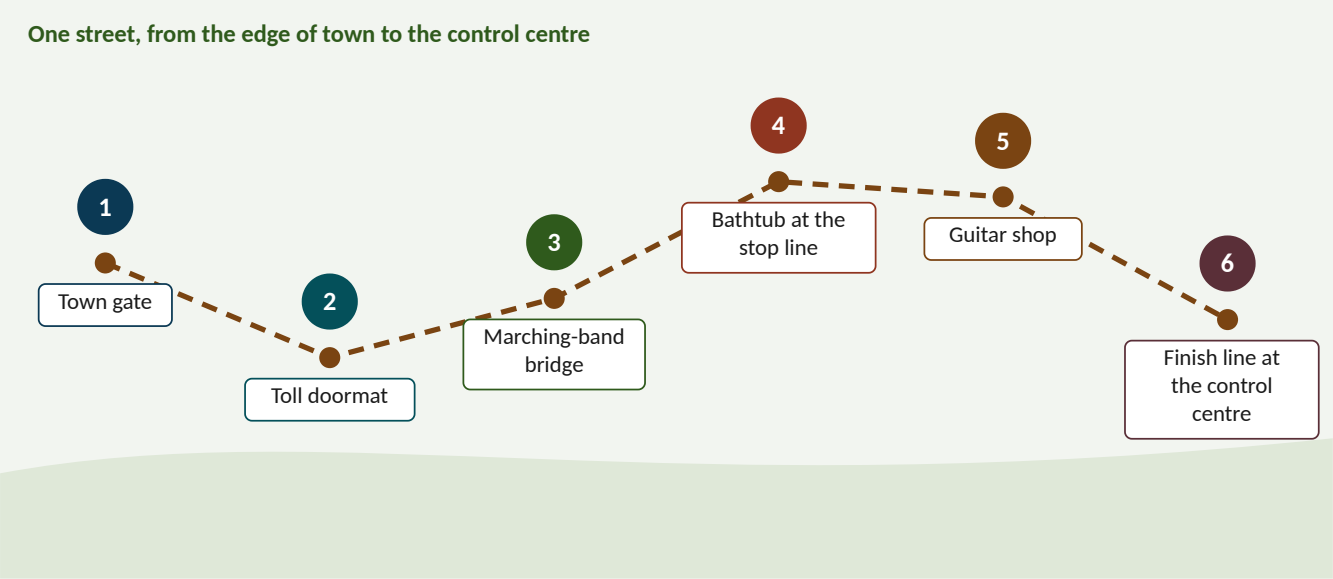
This…	…is not this	The one fact that tells them apart
Vehicles	**LPUs**	LPUs add occupancy, so a car is about 2 LPU and a bus about 3.
Cruise time T	**Lag βT**	The platoon's front arrives at 0.8 T, not T.
Displayed green	**Effective green**	Effective green starts a start lag late and runs an end lag into amber.
DoS 100%	**Gridlock**	100% means the green is just used up; queues grow only above it.
Elastic plan	**Plan selection**	SCOOT edits the running plan; plan selection swaps whole stored plans.

Symbols and terms

Symbol / term	Meaning
LPU	link profile unit: count plus occupancy, this lab uses 1·veh + 2·s
T	cruise time from detector to stop line (s)
α, β	Robertson's dispersion constants, 0.35 and 0.8
F	smoothing factor $1/(1 + \alpha\beta T)$
SATO	saturation occupancy: queue discharge rate in LPU/s
DoS	degree of saturation: demand ÷ (effective green × SATO)
PI	performance index = delay + K × stops
L, Y, C_0	Webster's lost time, sum of flow ratios, optimum cycle

The route: One street, from the edge of town to the control centre

Picture one long street you know. You walk it from the edge of town towards the traffic control centre. Six things happen along the way, one for each concept. Make every scene loud, silly and physical.



The memory route. Walk it in order; each stop holds one concept.

Stop 1: Town gate (What SCOOT is: the elastic plan). At the town gate a giant rubber band is stretched around the traffic light. Three dials on the pole are labelled S, C and O. A tiny engineer twangs the band every few seconds; it stretches, never snaps. *The rubber band is the elastic plan; the dials are split, cycle and offset; the frequent twangs are small, frequent changes.*

Stop 2: Toll doormat (Detectors and link profile units). Just past the gate, a doormat across the whole road is a cash register. Every vehicle pays ONE gold coin to roll on, then TWO coins for every second it sits there. A long red bus groans, paying three coins and dropping one on its foot. *LPU = 1 × vehicles + 2 × seconds occupied; the bus costs about 3 LPUs, a car about 2. The mat is far from the lights: the upstream detector.*

Stop 3: Marching-band bridge (Profiles and platoon dispersion). On a bridge, a marching band of 35 players leaves in a tight block. A sign says 'walk at 80% speed'. By the far end of the bridge the trombones are way ahead and the drummers trail far behind, still playing. *35 players and 80% encode $\alpha = 0.35$ and $\beta = 0.8$; the band spreading out is dispersion; the far end of the bridge is the stop line.*

Stop 4: Bathtub at the stop line (The queue model and degree of saturation). At the stop line stands a huge bathtub with a tap that never stops. When the light is red, a rubber plug shaped like a red light is in. On green the plug pops and water swirls out. A gauge on the side is painted red above the 90% mark. *Tap = arrivals; drain speed = SATO; red plug = red; the 90% mark = SCOOT's DoS target; the water's area over time = delay.*

Stop 5: Guitar shop (Validation: making the model match the street). Beside the stop line a guitar shop has a giant guitar strung across the road. An engineer with a stopwatch in her teeth twists a tuning peg marked SATO. The peg has two smaller pegs labelled 3 and 4. *Tuning = validation; the SATO peg = saturation occupancy; 3 and 4 = start lag 3 s and end lag 4 s; the stopwatch = observed queue clear time.*

Stop 6: Finish line at the control centre (SCOOT against fixed time and actuation). Outside the control centre three runners finish a race. A robot with a printed timetable on its chest crosses at 19, a jumpy rabbit at 14 and a half, and a calm dispatcher wearing headphones at just under 13. On the podium sits a set of scales labelled '1.5 L + 5'. *Finish times are the Lab F3 delays (19.0, 14.5, 12.8 s); the scales are Webster's formula* $C_0 = (1.5L + 5)/(1 - Y)$.

> **WALK-THROUGH DRILL**
>
> Close the guide. Walk the route in your head, saying each stop out loud and the one number or formula it holds. Then check against the cheat sheet. Repeat until you can do it in under two minutes.

Self-quiz

18 questions: warm-up (recall), core (use the ideas), challenge (combine them). Answer on paper first, then check the key.

Warm-up

Q1. What do the letters S, C and O in SCOOT stand for?

A. Speed, Capacity, Occupancy
B. Split, Cycle, Offset
C. Signal, Control, Optimisation
D. Stage, Cycle, Output

Q2. Where does SCOOT normally place its detectors?

A. At the stop line
B. In the middle of the junction
C. At the upstream end of each link
D. On the traffic light pole

Q3. How often does SCOOT normally review the region's cycle time?

A. Every second
B. Every stage change
C. Every cycle
D. Every 5 minutes (2.5 when rising)

Q4. What is SCOOT's usual target degree of saturation?

A. About 50%
B. About 90%
C. Exactly 100%
D. About 120%

Q5. Who stands at the stop line with a stopwatch during commissioning, and why?

A. A police officer, to fine red-light runners
B. An engineer, to validate the model
C. A driver, to time the cycle
D. No one, SCOOT calibrates itself fully

Core

Q6. A 7 m van crosses a 2 m loop at 13.5 m/s. Using this lab's LPU rule, how many LPUs?

A. 1.00
B. 1.67
C. 2.33
D. 3.07

Q7. Why does a slow car produce more LPUs than a fast car of the same length?

A. It is counted twice
B. It covers the loop for longer
C. SCOOT adds a penalty for slow cars
D. It does not; LPUs depend only on length

Q8. For cruise time T = 15 s, what is Robertson's F?

A. 0.192
B. 0.263
C. 0.350
D. 0.800

Q9. Dispersion changes a platoon between detector and stop line. Which statement is true?

A. It loses vehicles on the way
B. Its peak gets higher
C. Its total stays the same but it spreads out
D. It arrives exactly T seconds later, unchanged

Q10. Arrivals 0.25 LPU/s, effective green 30 s, SATO 0.5 LPU/s, cycle 60 s, no residual queue. DoS?

A. 0.50
B. 0.83
C. 1.00
D. 1.20

Q11. In the queue model, what is the delay in one cycle?

A. The longest queue
B. The number of stops
C. The area under the queue curve
D. The red time

Q12. The model's queue clears 5 s earlier than the real queue. What should the engineer do?

A. Raise SATO
B. Lower SATO
C. Raise the target DoS
D. Move the detector

Q13. Webster with L = 10 s and Y = 0.5. What is C_0?

A. 25 s
B. 40 s
C. 50 s
D. 80 s

Q14. In Lab F3 which controller had the lowest mean delay?

A. Fixed time
B. Actuated
C. SCOOT
D. All the same

Challenge

Q15. Two links: A has DoS 0.95, B has DoS 0.60, in a two-stage junction. What would a sensible split change do?

A. Give A's stage a few seconds from B's stage
B. Give B's stage more green
C. Lengthen the cycle by 60 s
D. Change nothing; both are under 100%

Q16. Why is an upstream detector better than a stop-line detector for predicting a queue?

A. It is cheaper
B. It sees vehicles before they reach the queue, so the model can predict
C. It counts more accurately
D. It is not; stop-line is always better

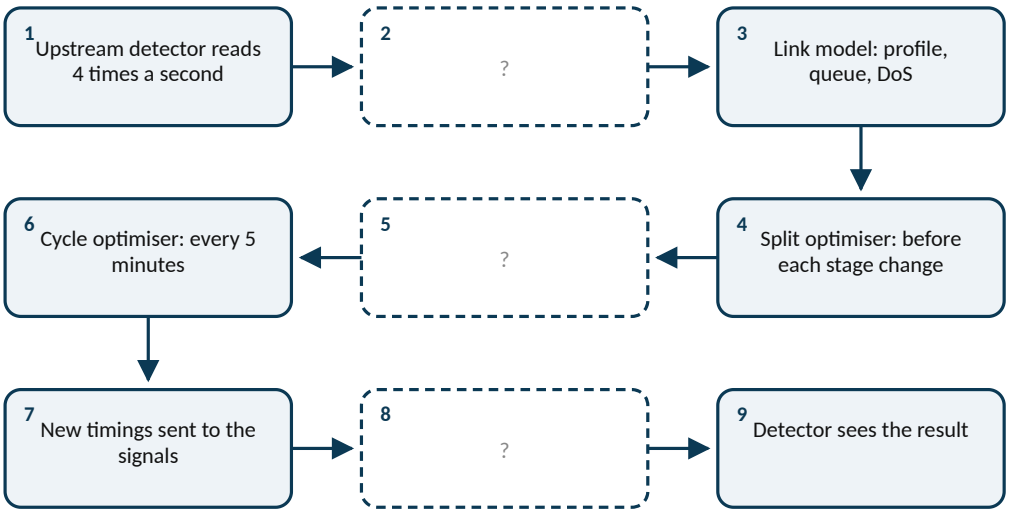
Q17. A road's cruise time doubles because a speed limit drops. What happens to F and the platoon?

A. F rises; platoon tighter
B. F falls; platoon spreads more
C. F unchanged; only lag changes
D. F becomes negative

Q18. Webster with Y = 0.95 and L = 10 gives C_0 = 400 s. What is the right conclusion?

A. Use a 400 s cycle
B. The junction is near capacity; cap the cycle and look at capacity, not timing
C. Webster is wrong for all junctions
D. Set Y to 0.6

Draw it. Fill the blank boxes of this flow from memory (it is Figure 1).



Where to go if you missed a question

Missed	Do this
Q1, Q3	Re-read concept 1: What SCOOT is: the elastic plan
Q2, Q6, Q7, Q16	Re-read concept 2: Detectors and link profile units
Q8, Q9, Q17	Re-read concept 3: Profiles and platoon dispersion
Q4, Q10, Q11, Q15	Re-read concept 4: The queue model and degree of saturation
Q5, Q12	Re-read concept 5: Validation: making the model match the street
Q13, Q14, Q18	Re-read concept 6: SCOOT against fixed time and actuation

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 min)

On paper, make a 60 s cycle on one link. Choose arrivals 0.15 LPU/s, red 30 s, green 30 s, SATO 0.5 LPU/s. Draw the queue, shade the delay, compute the DoS. Then shift 5 s of green away and redo it.
Self-check: With the first timing, the queue peaks at 4.5 LPU, clears 12.9 s into green and DoS = 9 ÷ 15 = 0.60. With 25 s of green, DoS = 9 ÷ 12.5 = 0.72.

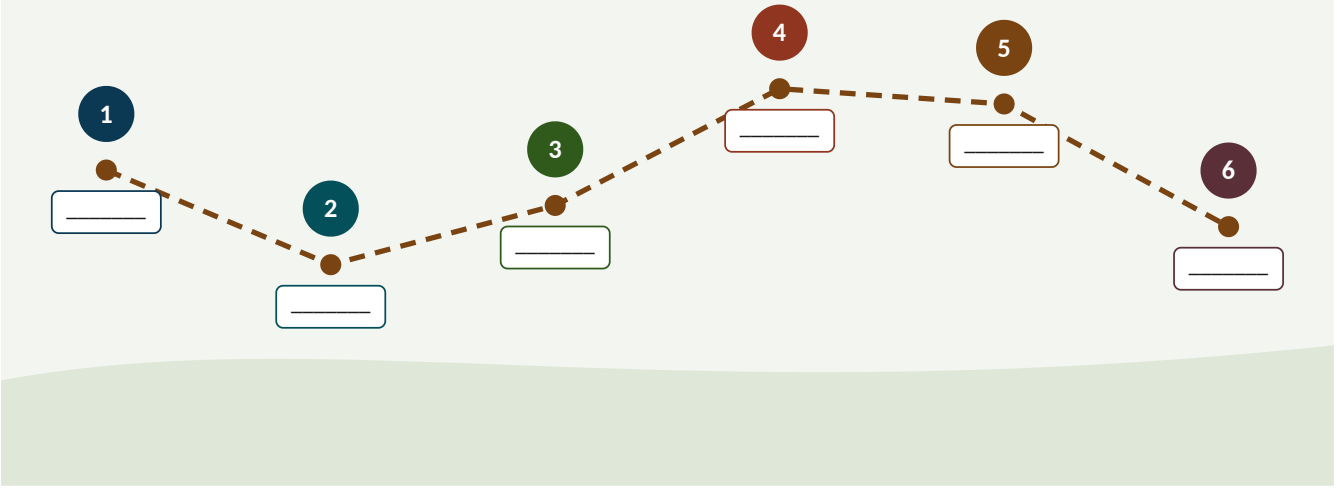
TEACH IT TO A 14-YEAR-OLD (10 min)

Explain to a 14-year-old how a traffic light could know a group of cars is coming before they get there, and why it changes its timing only a little at a time.
Words you may not use: LPU, saturation, optimiser, dispersion, occupancy

Fill the blank route

Write the concept held at each stop, and its key number.

One street, from the edge of town to the control centre



One-page cheat sheet

1. What SCOOT is: the elastic plan

Split = share of green; cycle = full round; offset = start time vs neighbour. Split every stage change, offset every cycle, cycle every 5 min (2.5 rising). Region shares one cycle.

Route: Town gate

2. Detectors and link profile units

Loop sampled 4×/s, placed upstream. Occupancy = (L_{veh} + L_{loop}) ÷ v. LPU = 1·veh + 2·s occupied. Car ≈ 2 LPU, bus ≈ 3.

Route: Toll doormat

3. Profiles and platoon dispersion

Profile smoothing 0.75 old + 0.25 new. $F = 1/(1+\alpha\beta T)$, α = 0.35, β = 0.8. Lag βT. T = 10 → F = 0.263, lag 8 s. T = 20 → F = 0.152, lag 16 s.

Route: Marching-band bridge

4. The queue model and degree of saturation

Effective green = green − start lag + end lag. Delay = area under queue. DoS = (arrivals + residual)/(g_{eff} × SATO), target 0.9. PI = delay + K·stops. Example: 0.2 LPU/s, r 35, g 25, SATO 0.6 → queue 7, delay 183.75, DoS 0.80.

Route: Bathtub at the stop line

5. Validation: making the model match the street

Validate: cruise time, SATO, start lag, end lag, max queue clear time. Clear time = start lag + Q/SATO. Model clears early → SATO too high. Lab F2: 1.95 LPU/veh, 13.5 m/s, SATO 0.67 main / 0.89 side, lags 3 s / 4 s.

Route: Guitar shop

6. SCOOT against fixed time and actuation

Webster $C_0 = (1.5L+5)/(1-Y)$; $g_i = (C_0-L)\cdot y_i/Y$. L 10, Y 0.6 → 50 s. SCOOT ≈ on-line TRANSYT. Lab F3 delay: fixed 18.96, actuated 14.48, SCOOT 12.79 s.

Route: Finish line at the control centre

Numbers to remember

- Loop read 4 times a second; 10-20 s warning
- Car ≈ 1.96 LPU, bus ≈ 3.07 LPU (13.5 m/s, 2 m loop)
- α = 0.35, β = 0.8; T = 10 → F = 0.263
- DoS target ≈ 90%
- Lab F2: SATO 0.67 / 0.89 LPU/s per lane, lags 3 / 4 s
- Lab F3 delay: 18.96 / 14.48 / 12.79 s
- Webster: L 10, Y 0.6 → C_0 = 50 s

Answer key

Try it now

Try it 1 (What SCOOT is: the elastic plan): 3600 ÷ 80 = 45 cycles; 3 stage changes per cycle gives 45 × 3 = **135** split decisions.

Try it 2 (Detectors and link profile units): Occupancy = (16.5 + 2) ÷ 9.25 = 18.5 ÷ 9.25 = 2.0 s. LPU = 1 + 2 × 2.0 = **5.0 LPU**. Slower vehicles cover the loop longer, so they score more too.

Try it 3 (Profiles and platoon dispersion): F = 1 ÷ (1 + 0.35 × 0.8 × 20) = 1 ÷ 6.6 = **0.152**; lag = 0.8 × 20 = **16 s**. F is smaller, so the platoon spreads **more**.

Try it 4 (The queue model and degree of saturation): Demand = 0.24 × 60 = 14.4 LPU; capacity 15 LPU; DoS = 14.4 ÷ 15 = **0.96**. That is **above** the 90% target, so the split optimiser would want more green here.

Try it 5 (Validation: making the model match the street): The model clears **later** than the street, so SATO is too **low**. SATO = 18 ÷ (27 − 3) = 18 ÷ 24 = **0.75 LPU/s**.

Try it 6 (SCOOT against fixed time and actuation): Y = 0.70. C_0 = (18 + 5) ÷ 0.30 = **76.7 s**. Green to share = 64.7 s. Main 64.7 × 0.40/0.70 = **37.0 s**; side 64.7 × 0.30/0.70 = **27.7 s**.

Self-quiz

Q1: B. Split Cycle Offset Optimisation Technique. The other options mix in real traffic words that are not the knobs SCOOT turns. (Concept 1)

Q2: C. Upstream placement gives 10-20 s of warning. Stop-line loops are typical of actuated control. (Concept 2)

Q3: D. Splits change every stage change and offsets every cycle; the cycle every 5 minutes. (Concept 1)

Q4: B. About 90% leaves a margin for random surges; 100% means the green is just full. (Concept 4)

Q5: B. Validation sets cruise time, SATO, lags and max queue clear time so the model matches the street. (Concept 5)

Q6: C. Occupancy 9 ÷ 13.5 = 0.667 s; LPU = 1 + 2 × 0.667 = 2.33. 3.07 is the bus; 1.00 forgets occupancy. (Concept 2)

Q7: B. Occupancy time = (length + loop) ÷ speed, so lower speed means more seconds occupied. (Concept 2)

Q8: A. F = 1 ÷ (1 + 0.35 × 0.8 × 15) = 1 ÷ 5.2 = 0.192. 0.263 is for T = 10; 0.35 and 0.8 are the constants themselves. (Concept 3)

Q9: C. Robertson's recursion conserves the total; it lowers and lengthens the peak. (Concept 3)

Q10: C. Demand 0.25 × 60 = 15 LPU; capacity 30 × 0.5 = 15 LPU; DoS = 1.00. The green is exactly full. (Concept 4)

Q11: C. Each LPU in the queue waits each second; summing gives the area (LPU·s). (Concept 4)

Q12: B. Clearing too early means the model thinks the queue leaves too fast: SATO is too high. (Concept 5)

Q13: B. C_0 = 20 ÷ 0.5 = 40 s. 50 s is the Y = 0.6 example. (Concept 6)

Q14: C. SCOOT 12.79 s, actuated 14.48 s, fixed 18.96 s. (Concept 6)

Q15: A. A is above the 90% target and B has spare green, so moving a few seconds from B to A balances them. Big cycle jumps are not SCOOT's style. (Concept 4)

Q16: B. Stop-line loops are covered by the standing queue and see arrivals too late; upstream loops give warning. (Concept 2)

Q17: B. F = 1/(1 + αβT) gets smaller as T grows, so arrivals spread more (and the lag βT also doubles). (Concept 3)

Q18: B. As Y → 1 no cycle can serve demand with reasonable delay; controllers cap the cycle (SCOOT's region maximum). (Concept 6)

Draw it: the completed flow is Figure 1.

Your revision plan

When	What	Done
Today	Read one concept a sitting. Do every Try-it-now on paper.	☐
Day 1	Walk the route from memory. Redraw the master mind map.	☐
Day 3	Warm-up and core quiz without notes. Re-read what you missed.	☐
Day 7	Whole quiz. Do the 'build the smallest example' task.	☐
Day 14	Teach-it task out loud. Cheat sheet from memory, then compare.	☐
Before an exam or project	Cheat sheet plus route walk only.	☐

Going further

- Hunt, P.B., Robertson, D.I., Bretherton, R.D. and Winton, R.I. (1981). *SCOOT: a traffic responsive method of coordinating signals*. TRRL Laboratory Report LR1014.
- Robertson, D.I. (1969). *TRANSYT: a traffic network study tool*. RRL Report LR253.
- Webster, F.V. (1958). *Traffic signal settings*. Road Research Technical Paper No. 39.

- This lab's website: Foundation page and SUMO labs F1-F3, smrahmanbd.github.io/scoot-learning-lab.

HONESTY NOTE

Worked examples use small, invented numbers chosen to be easy by hand. The LPU rule (1 × vehicles + 2 × seconds) is this lab's teaching definition; TRL's exact coding is not published. Lab F1-F3 results come from this repository's SUMO runs (seed 1) and are illustrative of the method, not of any real SCOOT installation.